Derivatives: Contracts, Models, and Markets · Technical Note 30

Two Adjustments Done Exactly: the Timing Covariance and the Quanto Density

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Chapter 30 of Derivatives: Contracts, Models, and Markets reduces every adjustment to one calculation: name the measure under which a variable's forward value is unbiased, name the measure the payoff requires, write the density between them, and take the covariance. This note takes two of the five cases and carries that calculation out without the shortcuts the chapter allows itself. Part I removes the linearization from the timing adjustment: the exact result is a covariance with the discount factor *rather than with the rate, the dropped term is a co-skewness, and the exponential form the market quotes turns out to be exact rather than cosmetic when the discount factor is lognormal. It is also quantified—on the chapter's own example the first-order formula understates the adjustment by 6.5%, and the error runs one way. Part II derives the quanto drift strictly by change of numeraire, exhibiting the density that Section 30.6 reaches around by Itô's product rule. That density is proportional to the* reciprocal *of the exchange rate, which is where the minus sign in $\mu_V = r_f - \rho\sigma_V\sigma_S$ comes from; getting the inversion backwards is the commonest error in the derivation, and a one-sentence sign check catches it.*

Part I. The timing adjustment without the linearization

1 THE EXACT STATEMENT

Work under Q^{T^*} and write

$$D = P(T^*, T), \qquad D_0 = \frac{P(0,T)}{P(0,T^*)} = \frac{1}{1 + y_0\tau}$$

for the discount factor over the deferral period $[T^*, T]$ and its forward value today. Because $P(t,T)/P(t,T^*)$ is a traded asset over the numeraire, $E^{T^*}[D] = D_0$ exactly, and the density of Chapter 30's equation (30.4) evaluated at T^* is

$$\Lambda = \frac{D}{D_0} = \frac{1 + y_0\tau}{1 + y\tau}, \qquad E^{T^*}[\Lambda] = 1$$

Therefore

$$E^{T}[V(T^*)] = E^{T^*}[V\Lambda] = \frac{E^{T^*}[VD]}{D_0} = V_0 + \frac{\mathrm{Cov}^{T^*}(V, D)}{D_0} \qquad \text{(TN30.1)}$$

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with no approximation whatever. **The covariance is with the discount factor, not with the rate.** This is Chapter 30's equation (30.6); what follows is what the chapter does not do with it.

2 WHERE THE APPROXIMATION ENTERS

Chapter 30 replaces $\mathrm{Cov}(V, D)$ by a covariance with the rate. Writing $\Delta = y - y_0$ and expanding $\Lambda = 1/(1+u)$ with $u = \tau\Delta/(1+y_0\tau)$,

$$\Lambda \;=\; 1 \;-\; \frac{\tau\Delta}{1+y_0\tau} \;+\; \frac{\tau^2\Delta^2}{(1+y_0\tau)^2} \;-\; \cdots$$

so that

$$E^T[V] - V_0 \;=\; \underbrace{-\frac{\tau}{1+y_0\tau}\,\mathrm{Cov}^{T^*}(V, y)}_{\text{Chapter 30's (30.7)}} \;+\; \frac{\tau^2}{(1+y_0\tau)^2}\,\mathrm{Cov}^{T^*}\left(V, \Delta^2\right) \;-\; \cdots \quad \text{(TN30.2)}$$

The dropped leading term is a *co-skewness*: the covariance of V with the *squared* rate move. It is second order in $\tau\sigma_y$, which is why the approximation is good for short deferrals and deteriorates as either the deferral or the rate volatility grows.

3 WHEN THE EXPONENTIAL FORM IS EXACT

Equation (30.8) of Chapter 30 is usually presented as a cosmetic re-expression of (30.7). It is more than that. If V and D are jointly lognormal under Q^{T^*} with $E[V] = V_0$ and $E[D] = D_0$, then $E[VD] = V_0 D_0 \exp(\rho_{VD}\,\sigma_V \sigma_D T^*)$, and (TN30.1) gives

$$\boxed{E^T[V(T^*)] \;=\; V_0 \exp(\rho_{VD}\,\sigma_V\,\sigma_D\,T^*)} \qquad \text{(TN30.3)}$$

exactly, where σ_D is the volatility of the discount factor and ρ_{VD} its correlation with V. Chapter 30's exponent is this one with the first-order translations

$$\frac{dD}{D} = -\frac{\tau\,dy}{1+y\tau} \quad\Longrightarrow\quad \sigma_D \approx \frac{y_0\tau}{1+y_0\tau}\,\sigma_y, \qquad \rho_{VD} \approx -\rho$$

which reproduce $-\rho\,\sigma_V\sigma_y y_0\tau T^*/(1+y_0\tau)$ term for term. So the exponential form is exact in the discount-factor parametrization, and the approximation lives entirely in expressing σ_D and ρ_{VD} through the rate's volatility and correlation.

4 HOW MUCH IT COSTS

Take the parameters of Chapter 30's first example: $V_0 = 4.00\%$, $\sigma_V = 20\%$, $y_0 = 4.00\%$, $\sigma_y = 20\%$, $\rho = 0.7$, $T^* = 5$, $\tau = 0.25$.

If the discount factor is lognormal. Then $\sigma_D = 0.0019802$, $\rho_{VD} = -0.7$, and (TN30.3) gives the exponent -0.00138614 and $E^T[V] = 3.9945\%$—identical to the chapter's answer to ten decimal places, because the chapter's formula *is* (TN30.3) under this assumption.

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If the rate is lognormal. Then $D = 1/(1 + y\tau)$ is not lognormal, and (TN30.1) must be integrated. Fixing the rate's median so that $E[D] = D_0$ holds exactly:

	Exact	Chapter's (30.8)
$E^T[V]$	3.9941%	3.9945%
Adjustment	−0.5928 bp	−0.5541 bp

The approximation understates the adjustment by 6.5% of itself—about 0.04 basis points on this trade.

How the gap grows. Holding $T^* = 5$, $\rho = 0.7$ and $\sigma_V = 20\%$:

σ_y	τ	Exact (bp)	(30.8) (bp)	Error
20%	0.25	−0.5928	−0.5541	−6.5%
20%	1.00	−2.2780	−2.1481	−5.7%
20%	2.00	−4.3332	−4.1267	−4.8%
40%	0.25	−1.2571	−1.1074	−11.9%
40%	1.00	−4.6843	−4.2846	−8.5%
40%	2.00	−8.6655	−8.2109	−5.2%
60%	0.25	−1.9529	−1.6599	−15.0%
60%	1.00	−6.9125	−6.4096	−7.3%
60%	2.00	−12.3730	−12.2529	−1.0%

The error is dominated by the rate's volatility rather than by the deferral length: at $\sigma_y = 60\%$ and a quarterly deferral the first-order formula misses a sixth of the adjustment, while at 20% it is within a few percent whatever τ is. The sign never changes—the approximation always understates—which matters because, as Section 30.7 observes, an omitted adjustment biases every affected trade in one direction. So does an understated one.

Part II. The quanto adjustment, strictly by change of numeraire

5 WHAT HAS TO BE SHOWN

The setting has two economies, domestic and foreign. Write $B_d(t)$ and $B_f(t)$ for the two money market accounts, each started at 1 and accruing at its own short rate, and $S(t)$ for the exchange rate in *units of domestic currency per unit of foreign currency*. Let $V(t)$ be the foreign-currency price of a traded foreign asset paying no income, with

$$\frac{dV}{V} = \mu_V\,dt + \sigma_V\,dW_V, \qquad \frac{dS}{S} = \mu_S\,dt + \sigma_S\,dW_S, \qquad dW_V\,dW_S = \rho\,dt \quad \text{(TN30.4)}$$

A quanto pays a function of $V(T)$ in domestic currency at a contractual exchange rate. Valuing it requires the law of V under the *domestic* risk-neutral measure Q^d, and what

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is known is its law under the *foreign* one, Q^f, where V is an ordinary traded asset. The claim to be established is

$$\mu_V = r_f - \rho\,\sigma_V\,\sigma_S \qquad \text{under } Q^d \tag{TN30.5}$$

6 THE TWO NUMERAIRES

A measure is named by its numeraire, so the first step is to say which asset each of the two measures divides by. Q^d is the measure associated with B_d: every domestic traded price divided by B_d is a Q^d-martingale.

Q^f needs one observation before it can be named in domestic terms. The foreign money market account is not a domestic asset, but *its domestic value is*: a domestic investor can buy foreign currency and roll it at the foreign short rate, and the position is worth

$$N(t) \;=\; S(t)\,B_f(t) \tag{TN30.6}$$

domestic units. N is therefore a traded domestic asset, strictly positive, and the measure associated with it is exactly Q^f: a foreign traded price X has domestic value XS, and

$$\frac{X(t)S(t)}{N(t)} = \frac{X(t)S(t)}{S(t)B_f(t)} = \frac{X(t)}{B_f(t)}$$

so the domestic-asset-over-N ratios are precisely the foreign-asset-over-B_f ratios that Q^f makes martingales. The foreign risk-neutral measure is the domestic measure associated with the numeraire SB_f. Nothing else about the foreign economy is needed.

7 THE DENSITY

The change-of-numeraire theorem states that for numeraires M and N, with Q^M and Q^N the measures they induce,

$$\left.\frac{dQ^M}{dQ^N}\right|_{\mathcal{F}_t} \;=\; \frac{M(t)\,/\,M(0)}{N(t)\,/\,N(0)} \tag{TN30.7}$$

Here the target measure is Q^d, so $M = B_d$, and the source is Q^f, so $N = SB_f$ by (TN30.6). With $B_d(0) = B_f(0) = 1$,

$$\boxed{\Lambda_t \;=\; \left.\frac{dQ^d}{dQ^f}\right|_{\mathcal{F}_t} \;=\; \frac{B_d(t)}{S(t)\,B_f(t)} \cdot S(0)} \tag{TN30.8}$$

Which way up this fraction goes is the whole derivation. Λ is proportional to $1/S$, not to S: moving *to* the domestic measure divides by the domestic value of the foreign account. The reciprocal is dQ^f/dQ^d, and using it produces (TN30.5) with the sign reversed. Section 9 gives the check that catches the error without redoing the algebra.

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8 GIRSANOV

Take the two short rates deterministic for the moment; Section 10 removes the assumption. Then B_d and B_f contribute no randomness and (TN30.8) reads

$$\Lambda_t = e^{(r_d - r_f)t}\,\frac{S(0)}{S(t)} \tag{TN30.9}$$

Under Q^f the exchange rate seen from the foreign side, $X = 1/S$ in units of foreign currency per unit of domestic, satisfies

$$\frac{dX}{X} = (r_f - r_d)\,dt + \sigma_S\,dW_X^f$$

by the same argument that gives S a drift of $r_d - r_f$ under Q^d: the domestic account is a foreign traded asset once converted. Substituting into (TN30.9),

$$\frac{d\Lambda}{\Lambda} = \sigma_S\,dW_X^f \tag{TN30.10}$$

and the drift has vanished, as it must—a Radon–Nikodym derivative is a martingale under the measure it starts from, and that cancellation is a check on (TN30.8) in its own right.

Girsanov's theorem now reads the drift change straight off the density's volatility. For any process Y, the Q^d drift of Y exceeds its Q^f drift by the instantaneous covariation of Y with Λ, per unit of each:

$$\mu_Y^d \;=\; \mu_Y^f \;+\; \frac{1}{Y\Lambda}\,\frac{d\langle Y, \Lambda\rangle}{dt} \tag{TN30.11}$$

Apply it to V. Under Q^f the foreign asset earns the foreign short rate, $\mu_V^f = r_f$. For the covariation, note that $X = 1/S$ moves against S, so dW_X^f and dW_V have correlation $-\rho$, and by (TN30.10)

$$\frac{1}{V\Lambda}\,\frac{d\langle V, \Lambda\rangle}{dt} = \sigma_V\,\sigma_S\,(-\rho) = -\rho\,\sigma_V\sigma_S$$

which with (TN30.11) gives (TN30.5):

$$\mu_V = r_f - \rho\,\sigma_V\,\sigma_S \qquad \text{under } Q^d$$

No product rule, no covered interest parity substitution, and no appeal to a hedging argument—only the numeraire, the theorem that turns it into a density, and Girsanov.

9 THE SIGN CHECK

The algebra can be trusted or not, but the sign can be settled in one sentence, and it is worth having because the inversion in (TN30.8) is the step that goes wrong.

Suppose $\rho > 0$: the foreign asset tends to be high exactly when foreign currency is expensive in domestic terms. A dealer short the quanto must hold the asset, and

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holds more of it as it rises—so buys foreign currency when it is dear. That is a cost, it recurs, and a cost to the hedger is a reduction in what the asset can be assumed to earn. The drift must therefore come out *below* r_f, which is what (TN30.5) says and what $\Lambda \propto 1/S$ delivers. Had the density been taken proportional to S, the covariation would have been $+\rho\sigma_V\sigma_S$ and the drift would have exceeded r_f, rewarding the dealer for a cost.

10 STOCHASTIC RATES

Nothing above required deterministic rates except the passage from (TN30.8) to (TN30.9). Taking logarithms in (TN30.8),

$$d\ln\Lambda = d\ln B_d - d\ln S - d\ln B_f$$

so the density's diffusion is $-\sigma_S\, dW_S$ plus whatever the two accounts contribute, and (TN30.11) picks up one covariance term for each. With deterministic accounts only the first survives and (TN30.5) is exact; with stochastic rates the quanto drift acquires covariances between the asset and each account, which is the same structure Part I meets as the timing adjustment. In practice these terms are second order against $\rho\sigma_V\sigma_S$ for equity quantos and are routinely dropped; for a quanto on a *rate*, they are not, and the full expression is what a cross-currency desk uses.

11 THE QUANTO FORWARD

Once the drift is known the forward follows in one line, which is why Chapter 30 states it immediately after. Under Q^d the asset grows at $\mu_V = r_f - \rho\sigma_V\sigma_S$, so

$$E^d[V(T)] = V(0)\, e^{(r_f - \rho\sigma_V\sigma_S)T}$$

while the ordinary foreign-currency forward is $F_0^f = V(0)e^{r_f T}$. Dividing,

$$F_0^{\text{quanto}} \;=\; F_0^f\, e^{-\rho\,\sigma_V\sigma_S\,T} \tag{TN30.12}$$

which is the chapter's result. A quanto option is then priced with Black's formula on F_0^{quanto}, discounted at the domestic rate—the measure change having done all of its work in the forward.

12 WHAT THE TWO PARTS HAVE IN COMMON

Part I starts from a density and asks what it costs to linearize it. Part II starts from a numeraire and asks which way up the density goes. Both are the same discipline: the adjustment is a covariance with an explicit object, and the object is worth writing down before it is approximated. Chapter 30's Itô argument for the quanto reaches (TN30.5) by imposing r_d as the drift of two domestic traded assets and solving—a verification, which confirms the answer without exhibiting what produces it. The density is what generalizes: name the numeraire a contract actually settles against, write (TN30.7), and the adjustment follows for contracts the chapter never mentions.

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